

Blockchain-Based Blue Carbon Registry and MRV System

Team: [Your Team Name]

PS Number:
SIH25038

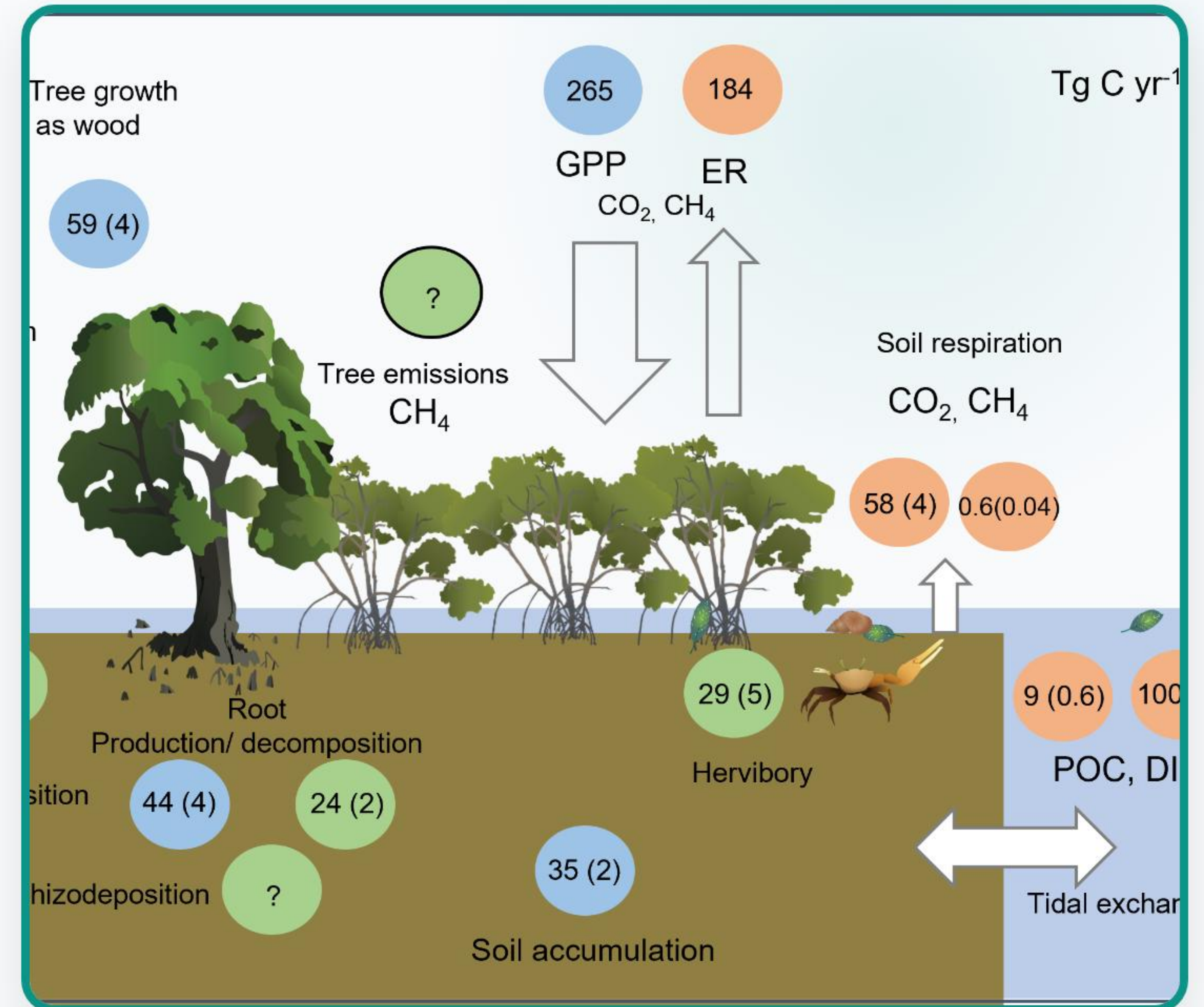
 Ministry of Earth
Sciences

 Theme: Clean & Green
Tech

 Category:
Software

Background: The Potential of Blue Carbon

- > **What is Blue Carbon?** Carbon captured by coastal ecosystems (mangroves, seagrasses, salt marshes).
- > Mangroves sequester carbon up to **4-5 times faster** than terrestrial forests.
- > **India's Coastal Assets:** ~5,000 sq km of mangroves, supported by national initiatives like the MISHTI scheme (2023).
- > **The Opportunity:** Restoring these ecosystems generates carbon credits that can be traded globally, supporting local livelihoods.
- > **The Core Need:** A verifiable, transparent system to quantify, track, and monetize these blue carbon credits for India.



The Current Challenge: Fragmented MRV



Manual & Error-Prone

Current Measurement, Reporting, and Verification (MRV) relies heavily on fragmented spreadsheets, isolated datasets, and manual field reporting.



Risk of Double-Counting

Without a centralized, immutable ledger, there is a significant risk of carbon credits being claimed or sold multiple times across different registries.

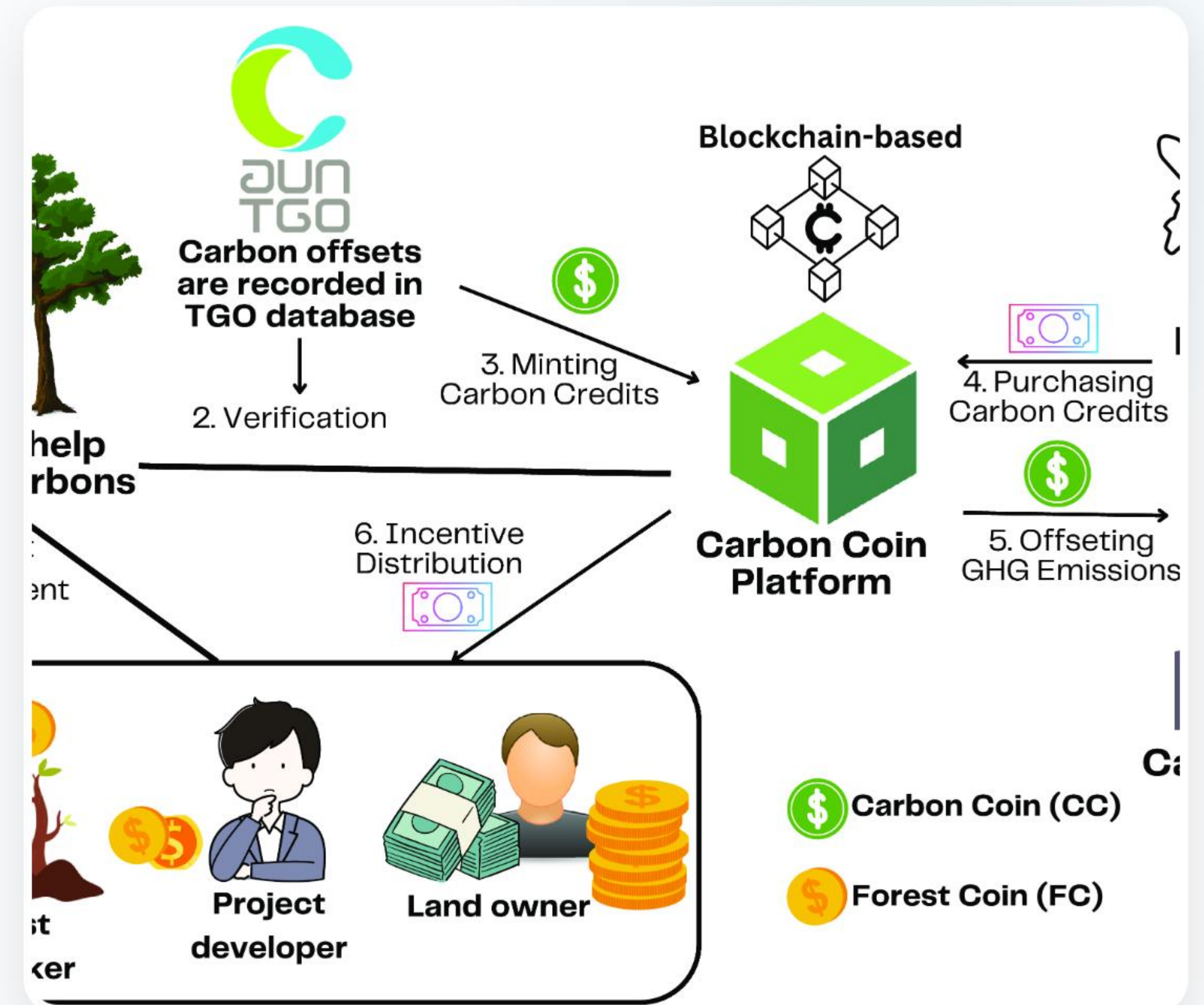


Stakeholder Friction

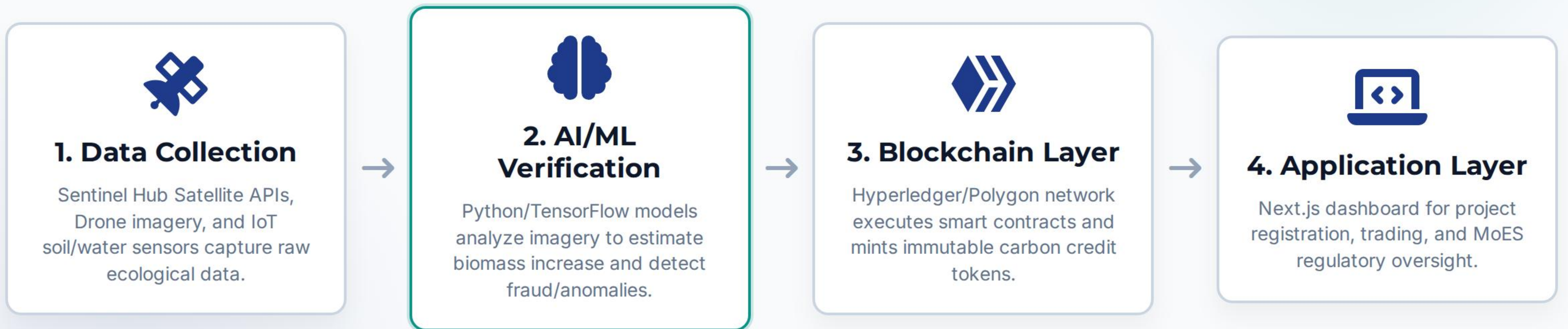
Lack of transparency makes it difficult to onboard local communities, NGOs, and panchayats, delaying the equitable distribution of financial incentives.

Our Solution: Decentralized Trust

- > **Digital Registry:** A unified platform to onboard blue carbon restoration projects with geo-tagged boundaries.
- > **Automated MRV:** Integrates IoT field sensor data and satellite imagery for continuous, remote monitoring.
- > **Tokenized Credits:** Issues verified carbon credits as immutable tokens (NFTs) on a blockchain to prevent double-spending.
- > **Smart Contracts:** Automates the verification workflow and credit issuance based on predefined ecological milestones.
- > **Public Dashboard:** Provides an auditable, transparent view for government bodies and



System Architecture & Data Flow



Proposed Technology Stack



Frontend

React.js / Next.js
Tailwind CSS



Backend & API

Node.js (Express) or Django
GraphQL



Blockchain

Hyperledger Fabric (Private)
or Polygon (Public)



Database & Storage

PostgreSQL
IPFS (for satellite images)



AI / ML Models

Python, TensorFlow
OpenCV



Geospatial Data

Sentinel Hub API
Google Earth Engine



Security & Auth

JWT Authentication
Role-Based Access



Deployment

Docker / Kubernetes
AWS / Azure

Feasibility & Mitigation Strategies

⚠️ Potential Challenges

- **Rural Connectivity:** Lack of internet in remote coastal/mangrove areas for real-time IoT updates.
- **Blockchain Gas Fees/Scalability:** High costs if using mainnet public blockchains for frequent transactions.
- **Standardization:** Aligning our MRV algorithms with global standards (Verra, Gold Standard).

Mitigation Plan

- **Offline-First Approach:** Mobile apps with local storage; sync data when network is available. Drone data processed locally.
- **Layer 2 / Permissioned Networks:** Use Polygon (L2) or Hyperledger Fabric for low-cost, high-throughput operations.
- **Modular API Design:** Build flexible APIs that map our MRV data outputs to Verra/Gold Standard reporting formats.

Projected Impact (Alignment with NDCs)

Env

Environmental Restoration

Accelerates tracking for India's target of restoring 540 sq km of mangroves under MISHTI, ensuring accurate carbon sequestration data.

Econ

Economic Growth

Unlocks access to the \$50B global voluntary carbon market for India. Ensures transparent revenue sharing directly to local panchayats.

Soc

Social Empowerment

Incentivizes coastal communities and fisherfolk to actively protect habitats rather than exploit them, creating sustainable "green jobs."

What Makes Our Approach Unique?



True Decentralization

Moves away from siloed, centralized government databases to a transparent ledger where NGOs, auditors, and buyers can independently verify claims.



Real-Time MRV

Replaces annual manual audits with continuous, automated monitoring using AI-analyzed satellite data, drastically reducing verification time.

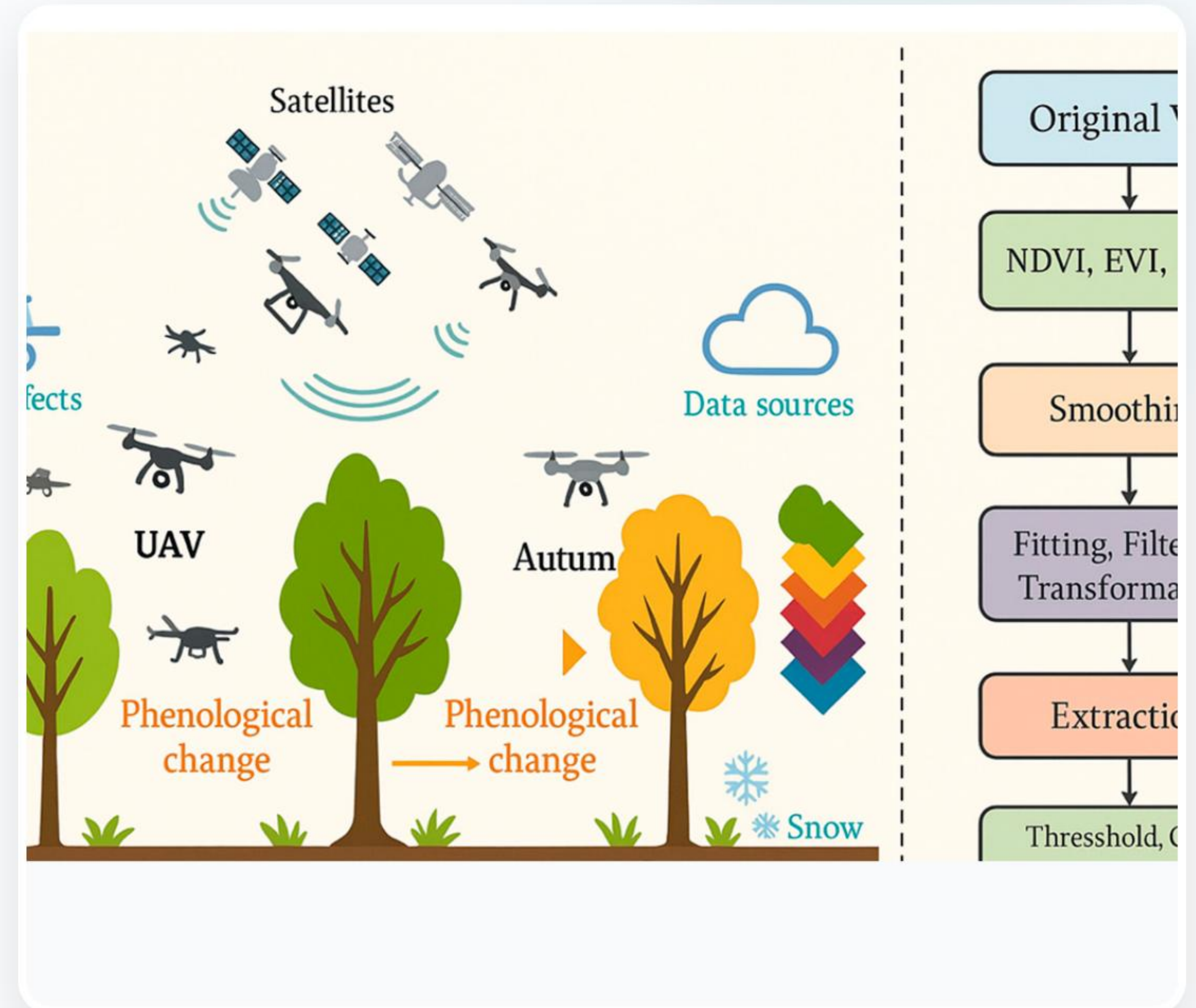


Community First

Smart contracts automatically trigger micropayments or fund disbursements to local communities based on verified ecological health metrics.

Integrating AI/ML for Verification

- > **Biomass Estimation Models:** Machine learning algorithms process multi-spectral satellite imagery (e.g., NDVI indices) to estimate mangrove canopy cover and carbon stock accurately.
- > **Change Detection:** Computer vision models compare historical and current drone/satellite imagery to track plantation growth or detect illegal deforestation.
- > **Fraud & Anomaly Detection:** AI flags inconsistencies between field-reported data and satellite observations before they are recorded on the blockchain.



Key Stakeholders & Beneficiaries



Regulators (MoES/MoEFCC)

Gain a unified, tamper-proof dashboard to track national blue carbon targets and enforce environmental compliance under schemes like MISHTI.



Coastal Communities

Direct financial beneficiaries of carbon credit sales. The system ensures fair, transparent, and prompt compensation for their conservation efforts.



Corporate Buyers

Access to high-quality, verified, and risk-free blue carbon credits to offset emissions and meet their ESG (Environmental, Social, Governance) goals.

Phase-Wise Implementation Roadmap

Phase 2: AI & MRV (M4-M6)

Integrate satellite AIs and train AI models for automated biomass estimation. Launch mobile app for field agents.

Phase 4: Market (M10-M12)

Integrate with international carbon exchanges. Enable corporate buying and automated smart-contract payouts.

Phase 1: Pilot (M1-M3)

Develop core blockchain ledger & web portal. Run a localized pilot in a single coastal district (e.g., Sundarbans or Gulf of Kutch).

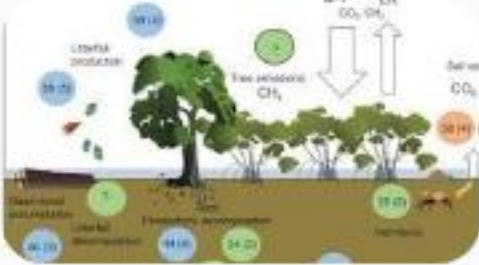
Phase 3: Scaling (M7-M9)

Expand deployment to multiple coastal states. Onboard NGOs and state forest departments to the platform.

References & Frameworks

- **MISHTI Scheme (2023):** MoEFCC initiative targeting 540 sq km of mangrove restoration across India.
- **National Coastal Mission:** Framework for conservation and management of mangroves and coral reefs.
- **India's NDCs:** Commitment to creating an additional carbon sink of 2.5 to 3 billion tonnes of CO₂ equivalent by 2030.
- **Global Standards:** Research aligned with Verra (VCS) and Gold Standard methodologies for wetland restoration.

CRZ, 2019: Salient Features			
Coastal Regulation Zone	Different Elements of the CRZ & its Features	Prohibited activities within the entire CRZ	Exceptions & regulation of permissible activities
Sea from HTL to 500 meters on land side Influenced water bodies - Land between HTL to 50 meters or the creek, whichever is less on land side along tidal-influenced areas connected to the sea up to where tidal effects are experienced Water & the bed area between from one bank to the LTL on the land side Land zone - land area b/w HTL & LTL Water & bed area between HTL & the territorial water limit (12 nautical miles) Backwater islands and islands along mainland coast: 20 meters from the landward side	1. CRZ-I: environmentally critical areas 1.1 CRZ-I A: ecologically sensitive areas 1.2 CRZ-I B: intertidal zone 2. CRZ-II: urban areas developed up to or close to the shoreline 3. CRZ-III: rural areas 3.1 CRZ-III A: population density of more than 2161 per square kilometre 3.1 CRZ-III B: population density of less than 2161 per square kilometre 4. CRZ-IV: water area 4.1 CRZ-IV A: water area and the sea bed between the Low Tide Line up to 12 nautical miles on the seaward side 4.2 CRZ-IV B: water & bed area b/w the low tide lines situated on the opposite banks of tidal influenced water bodies, extending from the mouth of the water body at sea up to where the tidal influence ends 5. Areas requiring special consideration - Critical coastal environments accorded special protection considering the difficulties faced by local communities 5.1 Critically Vulnerable Coastal Areas: (13 in number, egs. - Sunderbans, Gulf of Mannar, Vembanad, Bhitarkanika, etc.) 5.2 Inland Backwater islands and islands along the mainland coast 5.3 CRZ falling within municipal limits of Greater Mumbai	(i) New industries & expansion of existing industries, operations or processes. (ii) Manufacture or handling of oil, storage or disposal of hazardous substances (iii) New fish processing units. (iv) Land reclamation, bunding or disturbing the natural course of seawater (v) Discharging untreated wastes & effluents (vi) Dumping of city or town wastes for land-filling (vii) Port and harbour projects in high-eroding coastal stretches (viii) Mining of sand, rocks & other sub-strata materials (ix) Dressing or altering of active sand dunes (x) Disposing of plastic into coastal waters (xi) Withdrawing groundwater	1.1 CRZ -I A: Eco-tourism & roads etc. 1.2 CRZ-I B: Land reclamation, bunding for defence projects, projects for foreshore, erosion control, facilities for non-conventional energy sources a treatment, manual mining of atomic minerals, desalinization plants, etc. 2. CRZ-II: Activities permitted in CRZ-II. Vacant plots to be developed for tourism. 3. CRZ-III: Activities permitted in CRZ-III. Additionally, in CRZ-III, dispensaries, schools, colleges, toilets, bridges, etc. can be constructed. CRZ-III areas beyond the NDZ, mining of atomic minerals, with groundwater, etc., can occur. 4. CRZ-IV: Traditional fishing & allied activities by local communities, strategic defence projects including coastal network, exploration and extraction of natural gas, memorials, etc. 5. Areas requiring special consideration 5.1 Critically Vulnerable Coastal Areas (CVCAs): Integrated Management Plans (IMPs) prepared 5.2 Backwater islands and islands along mainland coast: Repairing or reconstructing existing units of local communities Fishing jetty, fish drying yards, net yards, traditional fishing processing boat building yards, ice plants, etc. States or UTs to prepare IIMPs for islands 5.3 CRZ areas within Greater Mumbai municipal limits: Sewage treatment plants Open spaces, parks, gardens, playgrounds within CRZ-II to be notified



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Source: wetlandinfo.detsi.qld.gov.au



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